[redacted] Ana 2008

Aufg 1)

1 Quadrant

a) $z = 3 + 2i$

$\tan \varphi = \frac{2}{3} = 33{,}7°$

$\|z\| = \sqrt{9+4} = \sqrt{13}$

$z = \sqrt{13}\, e^{i\, 33{,}7°}$

b)

$z = 3 e^{i \frac{\pi}{4}}$ $\qquad \varphi = \frac{\pi}{4} \hat{=} 45°$

[redacted] $3 \cdot \cos 45° + i \cdot 3 \sin 45° = 2{,}1 + i\, 2{,}1$

c)

$z = \frac{2+3i\,(1+i)}{(1-i)(1+i)} = \frac{2+2i+3i-3 \cdot i^2}{2} = \frac{-1+5i}{2}$

$z = 2+3i$ $\qquad$ 1Q $\tan \varphi = \frac{3}{2} = 56°$ $\qquad z \cdot (1-i)$ $\qquad$ 4 Quad. $\varphi = 315°$

$\|z\| = \sqrt{15}$ $\qquad \|z\| = \sqrt{2}$

$z = \sqrt{15}\, e^{i\, 56°}$ $\qquad z = \sqrt{2}\, e^{i\, 315}$

[redacted]

$\frac{z_1}{z_2} = \frac{\sqrt{15}\, e^{i\, 56°}}{\sqrt{2}\, e^{i\, 315}} = 2{,}7 \cos 101 + 2{,}7 \sin 101$

$\approx -0{,}5 + 2{,}5i$

[redacted] beide Ergebnisse sind gleich!

d) $z = i$ $\qquad r = \frac{\pi}{2}$

$\sqrt{1^2 + 0^2} = 1$

$z^{\frac{1}{2}} = 1^{\frac{1}{2}}\left(e^{i(\frac{\pi}{2} + m\pi)}\right)^{\frac{1}{2}} = 1^{\frac{1}{2}}\left(e^{i(\frac{\pi}{4} + \frac{m\pi}{2})}\right)$

$m = 0 \quad z^{\frac{1}{2}} = 1^{\frac{1}{2}}\left(e^{i\frac{\pi}{4}}\right.$

$m = 1 \quad z^{\frac{1}{2}} = 1^{\frac{1}{2}}\left(e^{i\frac{3}{4}\pi}\right.$ $\qquad \frac{\pi}{2}, \frac{3\pi}{2}$

$m = 2 \quad z^{\frac{1}{2}} = 1^{\frac{1}{2}}\left(e^{i\frac{5}{4}\pi}\right.$

$m = 3 \quad z^{\frac{1}{2}} = 1^{\frac{1}{2}}\left(e^{i\frac{7}{4}\pi}\right.$

Aufg 2 Am zeel

a) $\sum_{k=1}^{n} k^2 > \frac{n^3}{3}$

Ind Anf. $n=1$

$1 > \frac{1}{3}$ ✓

Ind Vor. $n=m$

Ind Sellw. $n=m+1$

$\sum_{k=1}^{m+1} k^2 > \frac{(m+1)^3}{3}$

$\sum_{k=1}^{m} k^2 + (m+1)^2 > \frac{(m+1)^3}{3}$

$\frac{(m)^3}{\cancel{3}} + 3(m+1)^2 > \frac{(m+1)^3}{\cancel{3}}$

$3(m^2+2m+1)+m^3 > m^3+3m^2+3m+1$

$m^3+3m^2+6m+6 > m^3+3m^2+3m+1$

$3m+5 > 0$ ✓

b) $(1+x)^n > 1+nx$

Ind Anf $n=2$

$(1+x)^2 > 1+2x$

$1+2x+x^2 > 1+2x$ ✓

Ind Vor $n=m$

Ind Schluss $n=m+1$

$(1+x)^{m+1} > 1+(m+1)x$

$(1+x)^m \cdot (1+x) > 1+(m+1)x$

$(1+mx)\cdot(1+x) > 1+mx+x$

$1+x+mx+mx^2 > 1+mx+x$

$mx^2 > 0$ ✓

Aufg 2c)

$0 < x < y$ Folgt $0 < x^n < y^n$

Ind [illegible] $n \to n+1$

$0 < x < y$ ✓

Ind [illegible] [illegible]

Ind Schluss [illegible]

$0 < x^{m+1} < y^{m+1}$

$0 < x^m < y^m$ [illegible]

Aufg 3 $y = \sin^2 x$

$y = \sin^2 x = \sin x \cdot \sin x$

$y' = \cos x \sin x + \sin x \cos x = 2\sin x \cos x$

$y'' = 2(\cos^2 x - \sin^2 x) = 2\cos^2 x - 2\sin^2 x$

$u = \sin x \qquad u' = \cos$

$v = \cos x \qquad v' = -\sin$

Aufg 4)

a) $y = x^{(x^2 \sin x)}$ $\qquad x = e^{\ln x}$

$y = e^{(\ln x)\, x^2 \sin x}$

$y' = e^{(\ln x)\, x^2 \sin x} \cdot 2x \sin x \cdot \cos x = e^{x^2 \sin x} \cdot 2x \sin \cdot \cos$

b) $y = \dfrac{\ln(\operatorname{tg} x)}{\sqrt{x}}$

$u = \ln(\operatorname{tg} x)$ $\qquad u' = \dfrac{1}{x \tan x}(1 + \tan^2 x) = \dfrac{\tan x}{x} + \tan^2 x$

$v = \sqrt{x}$ $\qquad v' = \dfrac{1}{2\sqrt{x}}$

$$y' = \frac{\left(\frac{\tan x}{x} + \tan^2 x\right) \cdot \sqrt{x} - \frac{\ln(\operatorname{tg} x)}{2\sqrt{x}}}{x}$$

c) $y = \operatorname{arctg} x$ $\qquad \operatorname{tg} y = x$ $\qquad$ — tan abgeleitet $\dfrac{1}{\cos^2}$

$\dfrac{dy}{dx} = \dfrac{1}{\frac{dx}{dy}}$ $\qquad \Rightarrow 1 = \dfrac{1}{\cos^2 y} \cdot y'$

$$\frac{1}{\frac{1}{\cos^2 y}} = \frac{1}{\frac{\cos^2 y + \sin^2 y}{\cos^2 y}} = \frac{1}{1 + \frac{\sin^2 y}{\cos^2 y}} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

ok $\tan y = \tan(\arctan) = x$

d) $\swarrow e^{-bx} \cdot \ln(ax)$

$$y = \frac{\ln ax}{e^{bx}}$$

$$y' = \frac{\frac{a}{x} e^{bx} - \ln ax \cdot e^{bx} b}{(e^{bx})^2}$$

$u = \ln ax \quad u' = \frac{a}{x}$

$v = e^{bx} \quad v' = e^{bx} \cdot b$

$$= \frac{\cancel{e^{bx}}}{(e^{bx})^{\cancel{2}}} \left(\frac{a}{x} - \ln ax \, b \right)$$

$$= \frac{1}{e^{bx}} \left(\frac{a}{x} - \ln ax \cdot b \right)$$

e)

$$y = \cos(\ln(\sin x))$$

$$y' = -\sin(\ln(\sin x)) \cdot \frac{1}{\sin x} \cdot \cos x$$

Aufg 5)

$u = \sin x \qquad du = \cos x \, dx$

a) $= \int \operatorname{ctg} x \, dx = \int \frac{\cos x}{\sin x} = \int \frac{du}{u} = \ln u + c$

$= \ln|\sin x| + c$

b) $= \int \arccos x \, dx = \int 1 \cdot \arccos x \, dx = x \cdot \arccos x + \int x \cdot \frac{1}{\sqrt{1-x^2}} \, dx$

$\underbrace{-\frac{1}{2}\int u^{\frac{1}{2}} du}_{= -u^{\frac{1}{2}}} \quad <= \quad \int x \cdot \frac{1}{\sqrt{1-x^2}}$

$x dx = d(\frac{1}{2}x^2)$

$= \frac{1}{2} dx^2 = -\frac{1}{2}(-x^2)$

$= -\frac{1}{2}(1-x^2)$

$= x \cdot \arccos x - \sqrt{1-x^2}$

c) $\int x \sin x \, dx = -x\cos x + \int \cos x$

$= -x\cos x + \sin x + c$

d) $\int x \ln x = \frac{1}{2}x^2 \ln x - \int \frac{1}{2}x^{\cancel{2}} \cdot \frac{1}{\cancel{x}}$

$= \frac{x^2}{2} \ln x - \frac{1}{2}\int x$

$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + c$

e) $\int e^{-\cos x} \sin x \, dx = \int e^{-\cos x} \, d(-\cos x)$

$-\cos x = u$

$\int e^u du = e^u = e^{-\cos x}$